

On the Non-decoupling of Supervised Fine-tuning and Reinforcement Learning in Post-training

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Abstract

Post-training of large language models routinely interleaves supervised fine-tuning (SFT) with reinforcement learning (RL). These two methods have different objectives: SFT minimizes the cross-entropy loss between model outputs and expert responses, while RL maximizes reward signals derived from human preferences or rule-based verifiers. Modern reasoning models have widely adopted the practice of alternating SFT and RL training. However, there is no theoretical account of whether they can be decoupled. We prove that decoupling is impossible in either order: (1) SFT-then-RL coupling: RL increases SFT loss under both distributional (KL-based) and landscape (PL-based) analyses; and (2) RL-then-SFT coupling: SFT lowers the reward achieved by RL under analogous conditions. Under the PL condition, we further derive the optimal RL duration that balances reward improvement against SFT degradation, identify the non-decoupling threshold governing when RL can improve SFT, and bound the gradient misalignment via spectral concentration. Experiments on Qwen3-0.6B confirm the predicted degradation, verifying that SFT and RL cannot be separated without loss of prior performance in the post-training pipeline.

1 Introduction

The capacity for reasoning and general tasks has been greatly improved in contemporary Large Language Models (LLMs) thanks to post-training techniques such as Supervised Fine-Tuning (SFT) and Reinforcement Learning (RL). Training LLMs typically comprises two stages, self-supervised *pretraining* and *post-training*, as illustrated in Figure 1. Test-time strategies can also be applied to improve the performance of post-trained models, but they do not modify the parameters of the post-trained model. During the pretraining stage, the model acquires general language patterns, structure, grammar, factual knowledge, and reasoning abilities by processing vast amounts of textual data. This stage of LLM training demands substantial computational resources. In addition, it involves extensive data cleaning to ensure that the model learns effectively and safely. Post-training often involves a combination of SFT and RL. SFT is a process that teaches the model how to respond to user prompts by providing task-specific input-output pairs, while the RL stage encourages certain patterns of text that have been positively reinforced by humans or rule-based verifiers. SFT has been shown to be prone to memorization, while RL is related to generalization (Chu et al., 2025; Huan et al., 2025). In this work, we study the synergy of

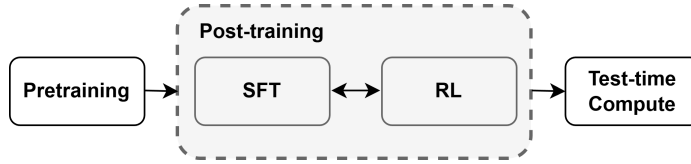


Figure 1: Training pipeline for modern LLMs. This work focuses on two post-training methods, Supervised Fine-Tuning (SFT) and Reinforcement Learning (RL), that refine a pretrained base model after its initial pretraining phase.

SFT and RL in the post-training pipeline, as shown in Figure 1, where pretrained models are further adapted with SFT and RL.

Modern reasoning models are built with alternating SFT and RL in practice. For example, DeepSeek-R1-Zero (Guo et al., 2025) is developed from DeepSeek-V3-Base using pure RL and has demonstrated remarkable reasoning capabilities. To further improve performance, DeepSeek-R1 (Guo et al., 2025) is post-trained from the DeepSeek-V3-Base model by alternating SFT and RL twice. The more recent post-training recipe (Wang et al., 2025) applies cascaded, domain-wise RL on top of a broad multi-stage SFT. Empirical evidence such as (Liu et al., 2025) suggests that a stronger SFT model consistently leads to better final performance after large-scale RL training. Notably, models often experience “catastrophic forgetting” during the transition from SFT to RL (Chen et al., 2025b). Liu et al. applied RL to a series of different SFT models and found that RL unlocks new abilities that the starting SFT models do not possess. However, Yan et al. showed that on-policy RL amplifies existing behaviors instead of introducing new capacities. In (Chen et al., 2025a), the authors proposed the “synergy dilemma” for large vision-language models by comparing three post-training strategies: two-stage SFT & RL, interleaved SFT & RL, and progressive SFT & RL. Their results show no synergy between SFT and RL in multi-modal reasoning. He et al. used a single RL stage to achieve state-of-the-art performance on two distilled 1.5B reasoning models. The diversity of techniques and their conflicting empirical outcomes raise these central questions:

**Does RL improve on prior SFT, and does SFT improve on prior RL?
Can the two post-training stages therefore be decoupled? To what
extent should we accept unavoidable capability degradation as the
cost of alignment, and when does that cost become unacceptable?**

In this paper, we approach these questions by analyzing two post-training pipelines: (a) SFT-then-RL, and (b) RL-then-SFT, as illustrated in Figure 2, as any cascade of the two strategies can be decomposed into these two elementary schemes. From the perspective of the SFT loss and the RL reward, we show that these two stages cannot be decoupled in terms of these objectives. In particular, our theoretical contributions are as follows.

1. **SFT-then-RL coupling:** In Theorem 3.1, we prove that when transitioning to RL training from SFT, the model performance in terms of SFT loss inevitably drops; therefore, SFT and RL cannot be decoupled in the SFT-then-RL pipeline.

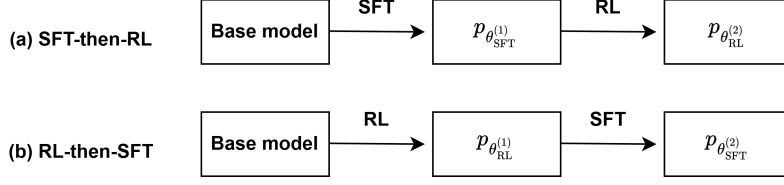


Figure 2: Any combination of SFT and RL in post-training reduces to the two canonical pipelines: (a) SFT-then-RL and (b) RL-then-SFT.

2. **RL-then-SFT coupling:** In Theorem 4.1, we prove that when transitioning to SFT from RL, the SFT step can create a persistent performance gap that decreases achieved RL reward. Therefore, SFT and RL interact non-trivially in the RL-then-SFT pipeline.
3. **Structural insights into the coupling:** We further characterize the coupling through three complementary results:
 - (a) Proposition 1 derives the optimal RL duration $T_{\text{RL}}^* = \lambda/(\mu_{\text{SFT}}\eta\bar{g}_{\text{RL}})$ under the Polyak–Łojasiewicz condition, showing that coupling damage grows quadratically while RL benefit grows linearly;
 - (b) Proposition 2 bounds the gradient misalignment $|\cos(g_{\text{SFT}}, g_{\text{RL}})|$ via spectral concentration in the eigenbasis, providing a mechanistic explanation for near-orthogonality; and
 - (c) Proposition 3 identifies the non-decoupling threshold τ_{crit} , giving the precise condition under which RL can (or cannot) improve SFT performance.

Empirically, we verify our Theorem 3.1 and Theorem 4.1 by conducting experiments on the Qwen3-0.6B model (Team, 2025). We post-train the model with the Corpus of Linguistic Acceptability (CoLA) dataset (Warstadt et al., 2019) using both SFT-then-RL and RL-then-SFT pipelines. Results show that RL diminishes SFT memory, as reflected by increased cross-entropy loss, and that RL becomes sensitive to further SFT, as shown by reward degradation. These paired changes indicate that the two training stages remain coupled and cannot be separated without loss of prior performance. Consequently, practitioners should treat SFT and RL as a single joint optimization problem rather than as separable blocks. In particular, our structural results suggest that the coupling damage scales quadratically with RL duration while reward improvement scales at most linearly, implying that there exists a theoretically optimal stopping point for RL training. Moreover, the near-orthogonality of SFT and RL gradients indicates that naive sequential optimization is fundamentally inefficient, so gradient surgery (Yu et al., 2020) or joint optimization strategies are needed to align the two objectives in parameter space.

1.1 Notation

Let $(\mathcal{X}, \mathcal{F}, \mu)$ be a measurable space with a fixed dominating measure μ . We use lower case p to denote the density of the corresponding probability measure P with respect to μ . A model corresponds to the probability distribution $p_\theta(\mathbf{x})$ parametrized by $\theta \in \Theta$. $\mathbf{x}_l^{l+m} := (x_l, x_{l+1}, \dots, x_{l+m})$ denotes a consecutive subsequence of tokens.

1.2 Preliminary

1.2.1 NEXT-TOKEN-PREDICTION LOSS

In general, a foundation model is pretrained with an autoregressive objective, followed by an SFT stage using labeled data. An autoregressive language model, parameterized by $\theta \in \Theta$, is a (discrete) distribution p_θ over $\mathcal{X}^*$ such that

$$p_\theta(\mathbf{x}) = \hat{p}(\text{EOS}|\mathbf{x}) \prod_{i=1}^L \hat{p}(x_i|\mathbf{x}_{<i}) \quad (1)$$

for $\mathbf{x} = (x_1, x_2, \dots, x_L) \in \mathcal{X}^*$ and $L = |\mathbf{x}|$, where the conditional distribution $\hat{p}(x|\mathbf{x})$ is a *sequence model* for $x \in \mathcal{X}$ and $\mathbf{x} \in \mathcal{X}^*$. During decoding, the next token is sampled according to a sampler $q(p_\theta)$ to generate sequences at the decoder that approximate the distribution of the language model. The model recursively generates each token until EOS to form a complete string.

Training is often conducted over a large corpus $\mathcal{D}$ using the *next-token-prediction* task by predicting the next token(s) $\mathbf{x}_i$ given a text chunk $\mathbf{x}_{<i}$. The objective is to minimize the negative log-likelihood

$$\mathcal{L}(\theta) := \frac{1}{|\mathcal{D}|} \sum_{\mathbf{x} \in \mathcal{D}} \sum_{i=1}^{|\mathbf{x}|} [-\log p_\theta(x_i | \mathbf{x}_{<i})]. \quad (2)$$

1.2.2 TV AND KL INEQUALITIES

Let p_X and p_Y be two probability distributions defined on the same σ -algebra $(\mathcal{X}, \mathcal{F})$, the TV distance is defined as

$$d_{\text{TV}}(p_X, p_Y) = \sup_{S \subseteq \mathcal{X}} |p_X(S) - p_Y(S)|,$$

which is equivalent to the L_1 norm when the alphabets are finite, i.e., when p_X and p_Y are probability mass functions,

$$d_{\text{TV}}(p_X, p_Y) = \frac{1}{2} \|p_X - p_Y\|_1 = \frac{1}{2} \sum_{x \in \mathcal{X}} |p_X(x) - p_Y(x)|. \quad (3)$$

As a metric on the space of probability measures, d_{TV} satisfies the triangle inequality.

$$d_{\text{TV}}(p_X, p_Z) \leq d_{\text{TV}}(p_X, p_Y) + d_{\text{TV}}(p_Y, p_Z). \quad (4)$$

A notable bound relates the expected values of bounded functions to the TV distance (see, e.g., [Niu et al. \(2025\)](#)):

$$|\mathbb{E}_{p_X}[f(X)] - \mathbb{E}_{p_Y}[f(X)]| \leq \sup_x |f(x)| d_{\text{TV}}(p_X, p_Y). \quad (5)$$

When $\mathcal{X}$ is finite, the KL divergence is defined as

$$D_{\text{KL}}(p_X \| p_Y) = \sum_{x \in \mathcal{X}} p_X(x) \log \frac{p_X(x)}{p_Y(x)}.$$

A fundamental relationship between the KL divergence and the TV distance, originally due to [Pinsker](#), states:

$$d_{\text{TV}}(P_X, P_Y) \leq \sqrt{\frac{1}{2} D_{\text{KL}}(P_X \| P_Y)}. \quad (6)$$

1.2.3 PL CONDITION

Our theoretical results rely on the Polyak-Łojasiewicz (PL) condition ([Polyak, 1963](#); [Karimi et al., 2016](#)). Here we discuss when this assumption is reasonable for neural network loss landscapes.

A function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ satisfies the PL condition with constant $\mu > 0$ if:

$$\frac{1}{2} \|\nabla f(\theta)\|^2 \geq \mu(f(\theta) - f^*), \quad \forall \theta \in \mathbb{R}^d \quad (7)$$

where $f^* = \inf_{\theta} f(\theta)$. This is weaker than strong convexity (which requires $\nabla^2 f \succeq \mu I$) but stronger than convexity. Importantly, PL does *not* require convexity, and non-convex functions can satisfy PL locally. In particular, PL is known to hold for over-parameterized neural networks ([Zhang et al., 2016](#); [Aich, 2025](#)).

2 Problem Setup

Let θ denote the parameters of a pretrained model p_{θ} . We consider two post-training strategies, SFT and RL, on performance evidenced by training loss and reward. We investigate the commutativity of the two sequential post-training operations, SFT-then-RL and RL-then-SFT, wherein SFT either precedes or follows RL, as illustrated in [Figure 2](#).

2.1 Supervised Fine-Tuning

In the SFT stage, the pretrained language model p_{θ} is trained with a next-token prediction loss using high-quality task-specific data to adapt to task-specific knowledge. Typically, SFT data consist of pairs of input prompts and desirable outputs $\mathcal{D}_{\text{SFT}} = \{(\mathbf{x}^i, \mathbf{y}^i)\}_{i=1}^{n_{\text{SFT}}} \sim p_{\mathcal{D}_{\text{SFT}}}$. The weights θ are updated to minimize the negative log-likelihood for each token of the anticipated output, which is equivalent to the cross-entropy loss

$$\mathcal{L}_{\text{SFT}}(p_{\theta}) := - \sum_{(\mathbf{x}, \mathbf{y}) \in \mathcal{D}_{\text{SFT}}} \sum_{j=1}^{|\mathbf{y}|} \log (p_{\theta}(\mathbf{y}_j \mid \mathbf{x}, \mathbf{y}_{<j})) \quad (8)$$

such that $\theta_{\text{SFT}} = \arg \min_{\theta} \mathcal{L}_{\text{SFT}}(p_{\theta})$. The resulting model $p_{\theta_{\text{SFT}}}(\mathbf{y}|\mathbf{x})$ generates $\mathbf{y}$ given a prompt $\mathbf{x}$. As the paradigm trains model to produce input-output mappings according to $\mathcal{D}_{\text{SFT}}$, it works well on in-distribution tasks, but may fail to generalize to out-of-distribution queries.

We give the following lemma, which states that the SFT loss can be equivalently expressed using chunks $\mathbf{x}$ and $\mathbf{y}$.

Lemma 1 *The autoregressive training loss (8) has the equivalent expression*

$$\mathcal{L}_{\text{SFT}}(\theta) := - \sum_{(\mathbf{x}, \mathbf{y}) \in \mathcal{D}_{\text{SFT}}} \log(p_{\theta}(\mathbf{y} | \mathbf{x})) = \mathbb{E}_{\mathbf{x} \sim p_{\mathcal{D}_{\text{SFT}}}(\mathbf{x}), \mathbf{y} \sim p_{\mathcal{D}_{\text{SFT}}}(\cdot | \mathbf{x})} [-\log p_{\theta}(\mathbf{y} | \mathbf{x})]$$

Proof

$$\begin{aligned} \mathcal{L}_{\text{SFT}}(\theta) &= - \sum_{(\mathbf{x}, \mathbf{y}) \in \mathcal{D}_{\text{SFT}}} \log \frac{1}{p_{\theta}(\mathbf{x})} \prod_{j=1}^{|\mathbf{y}|} p_{\theta}(\mathbf{x}) p_{\theta}(\mathbf{y}_j | \mathbf{x}, \mathbf{y}_{<j}) \\ &= - \sum_{(\mathbf{x}, \mathbf{y}) \in \mathcal{D}_{\text{SFT}}} \log \frac{1}{p_{\theta}(\mathbf{x})} \prod_{j=1}^{|\mathbf{y}|} p_{\theta}(\mathbf{y}_j | \mathbf{x}, \mathbf{y}_{<j}) \prod_{i=1}^{|\mathbf{x}|} p_{\theta}(\mathbf{x}_i | \mathbf{x}_{<i}) \\ &= - \sum_{(\mathbf{x}, \mathbf{y}) \in \mathcal{D}_{\text{SFT}}} \log(p_{\theta}(\mathbf{y} | \mathbf{x})) \end{aligned}$$

■

2.2 Reinforcement Learning

To align models with human preferences, the Reinforcement Learning from Human Feedback (RLHF) pipeline (Christiano et al., 2017; Ziegler et al., 2019; Ouyang et al., 2022) trains a reward model using preference data to encode desirable traits, then aligns the language model using policy gradient methods such as proximal policy optimization (PPO) (Schulman et al., 2017) and value-model-based RL such as PPO and variants DAPO (Yu et al., 2025), VAPO (Yue et al., 2025) to maximize reward. Reinforcement Learning with Verifiable Rewards (RLVR) (Lambert et al., 2025; Guo et al., 2025) is adopted when a clear reward function is possible while label data may be noisy or expensive. Instead of relying on human-labeled data, direct binary feedback (true/false) can be obtained from a formal verification tool, such as symbolic verifiers and rule-based tools.

Using standard RL terminology, the language model p_{θ} is referred to as a *policy*, where the reference model π_{ref} (often from the SFT stage) serves as the initial policy preceding the RL stages (see (Razin et al., 2025)). Similarly to SFT, which trains the model on input-output pairs, RL aims at producing a response y^* that takes the highest value on a reward function.

To unify RLHF and RLVR, we model the RL data $\mathcal{D}_{\text{RL}} = \{\mathbf{x}_i, \mathbf{y}_i^{\text{pos}}, \mathbf{y}_i^{\text{neg}}\}_{i=1}^{n_{\text{RL}}} \sim p_{\mathcal{D}_{\text{RL}}}(\mathbf{x}, \mathbf{y})$ as prompt-response pairs, where given prompt $\mathbf{x}$, $\mathbf{y}^{\text{pos}}$ and $\mathbf{y}^{\text{neg}}$ stand for positive (preferred/true) and negative responses annotated by humans or rule-based tools. An unknown ground truth reward

$$r_G(\cdot, \cdot) : \mathcal{X}^* \times \mathcal{X}^* \mapsto [-R, R]$$

is assumed to encode human preferences. When provided with an input $\mathbf{x}$, the reward $r_G(\mathbf{x}, \mathbf{y})$ evaluates the quality of the output $\mathbf{y}$. The objective is to maximize the expected reward

$$\max_{\theta} \mathbb{E}_{\mathbf{y} \sim p_{\theta}(\cdot | \mathbf{x})} [r_G(\mathbf{x}, \mathbf{y})].$$

When r_G is not directly accessible, many methods, e.g., (Ziegler et al., 2019; Bai et al., 2022; Ouyang et al., 2022), train a proxy reward model $r(\cdot, \cdot)$ through a Bradley-Terry log-likelihood loss (Bradley and Terry, 1952) on RL data $\mathcal{D}_{\text{RL}}$. The optimization is further regularized with entropy so that the model does not drift too far from the original model. The policy p_{θ} is updated using policy gradient methods (e.g. PPO, GRPO) to maximize the following RL objective

$$\mathcal{J}_{\text{RL}}(\theta) = \mathbb{E}_{\mathbf{x} \sim p_{\mathcal{D}_{\text{RL}}}, \mathbf{y} \sim p_{\theta}(\cdot | \mathbf{x})} [r(\mathbf{x}, \mathbf{y})] - \beta \mathbb{E}_{\mathbf{x} \sim p_{\mathcal{D}_{\text{RL}}}} [D_{\text{KL}}(p_{\theta}(\cdot | \mathbf{x}) \| \pi_{\text{ref}}(\cdot | \mathbf{x}))],$$

where the reward is bounded, $r(\mathbf{x}, \mathbf{y}) \leq R_{\text{max}}$, $p(x)$ is a fixed distribution over the prompts at test time, and $\theta_{\text{RL}} = \arg \max_{\theta} \mathcal{J}_{\text{RL}}(\theta)$. The closed-form solution is well known to be (e.g., see (Peng et al., 2019))

$$p_{\theta_{\text{RL}}} = \frac{1}{Z(\mathbf{x})} \pi_{\text{ref}}(\mathbf{y} | \mathbf{x}) \exp \left(\frac{r(\mathbf{x}, \mathbf{y})}{\beta} \right). \quad (9)$$

As β increases, the RL algorithm behaves more “on-policy” (Yan et al., 2025).

2.3 Assumptions

We make the following assumption regarding the SFT and RL data. In $\mathcal{D}_{\text{SFT}}$ and $\mathcal{D}_{\text{RL}}$, each sample consists of a pair $(\mathbf{x}_i, \mathbf{y}_i)$, where x is the input of the task, e.g., a prompt, a query, etc. This covers a wide range of scenarios, such as question-answering, summarization, and code completion. In particular, we assume that the prompts are from the same distributions, so that the SFT data and RL data are from the same domains.

Assumption 1 *The sets of prompts $\{\mathbf{x}_i\}$ in $\mathcal{D}_{\text{SFT}}$ and $\mathcal{D}_{\text{RL}}$ are sampled uniformly according to the prompt distribution $\mathbf{x} \sim q(\mathbf{x})$.*

In particular, Akter et al. studied the allocation of reasoning data between pretraining and different post-training stages, revealing that pretraining benefits from diversity in reasoning patterns and that SFT is sensitive to data quality.

We make the following bounded reward assumption during the RL stage, which is enforced in practice in standard algorithms such as PPO and GRPO.

Assumption 2 *Let $r(\cdot, \cdot)$ be the reward function during RL, then there exists $R_{\text{max}} > 0$ such that*

$$|r(\mathbf{x}, \mathbf{y})| \leq R_{\text{max}} \quad \forall (\mathbf{x}, \mathbf{y}) \in \mathcal{X}^{*2} \quad (10)$$

We also assume the following regularity condition on the SFT loss landscape, which is weaker than strong convexity and is known to hold for over-parameterized neural networks (Karimi et al., 2016; Aich, 2025).

Assumption 3 (Polyak–Łojasiewicz condition) *The SFT loss $\mathcal{L}_{SFT}$ satisfies the PL condition with constant $\mu_{SFT} > 0$ if*

$$\frac{1}{2} \|\nabla \mathcal{L}_{SFT}(\theta)\|^2 \geq \mu_{SFT} (\mathcal{L}_{SFT}(\theta) - \mathcal{L}_{SFT}^*)$$

for all θ , where $\mathcal{L}_{SFT}^* = \min_{\theta} \mathcal{L}_{SFT}(\theta)$.

3 SFT-then-RL Coupling

We first study the training loss of the SFT-then-RL pipeline illustrated in Figure 2(a). Let p_{θ} denote a pretrained base model upon which we perform SFT followed by RL. We denote the first SFT checkpoint as $p_{\theta_{SFT}^{(1)}}$ and the checkpoint obtained after the subsequent RL process as $p_{\theta_{RL}^{(2)}}$.

We use cross-entropy loss to indicate model performance, as in (Niu et al., 2024). The next theorem establishes a non-decoupling property: even when the model has already converged under the first SFT (so the SFT objective exhibits negligible further loss reduction), the subsequent RL phase can still impair the SFT-induced performance. In particular, SFT and RL cannot be decoupled: any nontrivial improvement in the RL reward necessarily induces a nontrivial degradation in the SFT loss.

In particular, suppose that the first SFT stage achieves a low loss (see Lemma 1), i.e.,

$$\mathcal{L}_{SFT}(p_{\theta_{SFT}^{(1)}}) = \mathbb{E}_{\mathbf{x} \sim q(\mathbf{x}), \mathbf{y} \sim p_{\mathcal{D}_{SFT}}(\cdot | \mathbf{x})} [-\log p_{\theta_{SFT}^{(1)}}(\mathbf{y} | \mathbf{x})] \leq \epsilon_{SFT}.$$

Then, after the subsequent RL stage, we show that the SFT loss *increases* by a positive amount:

$$\mathbb{E}_{\mathbf{x} \sim q(\mathbf{x}), \mathbf{y} \sim p_{\mathcal{D}_{SFT}}(\cdot | \mathbf{x})} [-\log p_{\theta_{RL}^{(2)}}(\mathbf{y} | \mathbf{x})] \geq \epsilon_{SFT} + \delta(\beta),$$

for some strictly positive $\delta(\beta)$ whenever the RL phase achieves nontrivial reward improvement. Consequently, the improvement attributable to RL is *not* orthogonal to the gains achieved during SFT: improving reward trades off against the SFT likelihood fit, so the two phases cannot be decoupled.

Theorem 3.1 *Suppose that the first SFT stage results in a reference model that matches the SFT data:*

$$p_{\theta_{SFT}^{(1)}}(\mathbf{y} | \mathbf{x}) = p_{\mathcal{D}_{SFT}}(\mathbf{y} | \mathbf{x}) \quad \text{for } q\text{-a.e. } \mathbf{x}, \forall \mathbf{y} \quad (11)$$

then the second RL phase degrades the SFT performance, i.e.

$$\mathbb{E}_{\mathbf{x} \sim q(\mathbf{x}), \mathbf{y} \sim p_{\mathcal{D}_{SFT}}(\cdot | \mathbf{x})} [-\log p_{\theta_{RL}^{(2)}}(\mathbf{y} | \mathbf{x})] = \mathbb{E}_{\mathbf{x} \sim q(\mathbf{x}), \mathbf{y} \sim p_{\mathcal{D}_{SFT}}(\cdot | \mathbf{x})} [-\log p_{\theta_{SFT}^{(1)}}(\mathbf{y} | \mathbf{x})] + C_1(\beta) \quad (12)$$

for some constant $C_1(\beta) \geq 0$.

Proof The SFT-then-RL scheme admits $\pi_{\text{ref}} = p_{\theta_{SFT}^{(1)}} \cdot \theta_{RL}^{(2)}$ further maximizes $\mathcal{J}_{RL}(\theta)$, and the maximizer is given by

$$p_{\theta_{RL}^{(2)}}(\mathbf{y} | \mathbf{x}) = \frac{1}{Z_{\beta}(\mathbf{x})} p_{\theta_{SFT}^{(1)}} \exp\left(\frac{r(\mathbf{x}, \mathbf{y})}{\beta}\right) \quad (13)$$

with $Z_\beta(\mathbf{x}) = \mathbb{E}_{\mathbf{y} \sim p_{\theta_{\text{SFT}}^{(1)}}(\cdot|\mathbf{x})} [\exp(r(\mathbf{x}, \mathbf{y})/\beta)]$ according to Equation (9). Using (13),

$$-\log p_{\theta_{\text{RL}}^{(2)}}(\mathbf{y} | \mathbf{x}) = -\log p_{\theta_{\text{SFT}}^{(1)}}(\mathbf{y} | \mathbf{x}) - \frac{1}{\beta} r(\mathbf{x}, \mathbf{y}) + \log Z_\beta(\mathbf{x}).$$

Taking expectations gives

$$\begin{aligned} \mathcal{L}_{\text{SFT}}(p_{\theta_{\text{RL}}^{(2)}}) &= \mathbb{E}_{\mathbf{x} \sim q, \mathbf{y} \sim p_{\mathcal{D}_{\text{SFT}}}(\cdot|\mathbf{x})} [-\log p_{\theta_{\text{SFT}}^{(1)}}(\mathbf{y} | \mathbf{x})] \\ &\quad - \frac{1}{\beta} \mathbb{E}_{\mathbf{x} \sim q, \mathbf{y} \sim p_{\mathcal{D}_{\text{SFT}}}(\cdot|\mathbf{x})} [r(\mathbf{x}, \mathbf{y})] + \mathbb{E}_{\mathbf{x} \sim q} [\log Z_\beta(\mathbf{x})] \\ &= \mathcal{L}_{\text{SFT}}(p_{\theta_{\text{SFT}}^{(1)}}) + C_1(\beta), \end{aligned} \tag{14}$$

where $C_1(\beta) = \mathbb{E}_{\mathbf{x} \sim q} [\log Z_\beta(\mathbf{x}) - \frac{1}{\beta} \mathbb{E}_{\mathbf{y} \sim p_{\mathcal{D}_{\text{SFT}}}(\cdot|\mathbf{x})} [r(\mathbf{x}, \mathbf{y})]]$.

Next, we show the nonnegativity of $C_1(\beta)$. For each fixed $\mathbf{x}$, apply Jensen's inequality to the convex function $\exp(\cdot)$:

$$\mathbb{E}_{\mathbf{y} \sim p_{\mathcal{D}_{\text{SFT}}}(\cdot|\mathbf{x})} [e^{r(\mathbf{x}, \mathbf{y})/\beta}] \geq \exp \left(\mathbb{E}_{\mathbf{y} \sim p_{\mathcal{D}_{\text{SFT}}}(\cdot|\mathbf{x})} [r(\mathbf{x}, \mathbf{y})]/\beta \right).$$

Taking logarithms on both sides and averaging over $\mathbf{x} \sim q$ yields $C_1(\beta) \geq 0$. ■

We also provide a complementary result under the PL condition (Assumption 3), which gives an explicit quadratic lower bound on the SFT loss increase in terms of parameter displacement.

Theorem 3.2 (PL-based SFT Degradation) *Let $\theta_{\text{SFT}}^{(1)} = \arg \min_{\theta} \mathcal{L}_{\text{SFT}}(\theta)$ be the SFT-optimal parameter. Suppose $\mathcal{L}_{\text{SFT}}$ satisfies the PL condition with constant $\mu_{\text{SFT}} > 0$. Then after RL optimization from this point, the resulting parameter $\theta_{\text{RL}}^{(2)}$ satisfies:*

$$\mathcal{L}_{\text{SFT}}(\theta_{\text{RL}}^{(2)}) - \mathcal{L}_{\text{SFT}}(\theta_{\text{SFT}}^{(1)}) \geq \frac{\mu_{\text{SFT}}}{2} \|\theta_{\text{RL}}^{(2)} - \theta_{\text{SFT}}^{(1)}\|^2. \tag{15}$$

The inequality is strict when $\nabla \mathcal{J}_{\text{RL}}(\theta_{\text{SFT}}^{(1)}) \neq 0$.

Proof By the PL condition, for any θ :

$$\mathcal{L}_{\text{SFT}}(\theta) - \mathcal{L}_{\text{SFT}}(\theta_{\text{SFT}}^{(1)}) \geq \frac{1}{2\mu_{\text{SFT}}} \|\nabla \mathcal{L}_{\text{SFT}}(\theta)\|^2 \geq 0,$$

since the PL condition implies quadratic growth. In particular, applying the quadratic growth property $\mathcal{L}_{\text{SFT}}(\theta) - \mathcal{L}_{\text{SFT}}^* \geq \frac{\mu_{\text{SFT}}}{2} \|\theta - \theta_{\text{SFT}}^{(1)}\|^2$ with $\theta = \theta_{\text{RL}}^{(2)}$ yields the result. When $\nabla \mathcal{J}_{\text{RL}}(\theta_{\text{SFT}}^{(1)}) \neq 0$, the RL optimizer must move away from $\theta_{\text{SFT}}^{(1)}$ to improve the reward, so $\theta_{\text{RL}}^{(2)} \neq \theta_{\text{SFT}}^{(1)}$ and the inequality is strict. ■

The quadratic dependence on $\|\theta_{\text{RL}}^{(2)} - \theta_{\text{SFT}}^{(1)}\|$ implies that the SFT degradation scales quadratically with RL duration, while the RL reward improvement scales at most linearly. This quadratic-versus-linear trade-off naturally leads to the question of optimal RL stopping, which we address in Proposition 1.

3.1 Optimal RL Duration

The SFT-then-RL coupling established in Theorem 3.1 raises a practical question: how long should RL training last before the cumulative SFT damage outweighs the reward improvement? Under Assumption 3, the SFT loss satisfies quadratic growth: $\mathcal{L}_{\text{SFT}}(\theta) - \mathcal{L}_{\text{SFT}}^* \geq \frac{\mu_{\text{SFT}}}{2} \|\theta - \theta_{\text{SFT}}^*\|^2$ for all θ (Karimi et al., 2016). This implies that any parameter displacement from the SFT optimum incurs at least quadratic SFT loss increase.

To analyze the trade-off between RL reward improvement and SFT degradation, we introduce the RL loss $\mathcal{L}_{\text{RL}}(\theta) := -\mathcal{J}_{\text{RL}}(\theta)$, so that minimizing $\mathcal{L}_{\text{RL}}$ is equivalent to maximizing the reward $\mathcal{J}_{\text{RL}}$.

Proposition 1 (Optimal RL Duration) *Let $\theta_{\text{SFT}}^{(1)}$ be the SFT optimum after T_{SFT} steps, with PL constant μ_{SFT} . Assume RL runs for T_{RL} steps with learning rate η and mean RL gradient norm $\bar{g}_{\text{RL}} = \frac{1}{T_{\text{RL}}} \sum_{t=1}^{T_{\text{RL}}} \|g_{\text{RL}}^{(t)}\|$. Define the coupling coefficient $\kappa = \mu_{\text{SFT}} \eta^2 \bar{g}_{\text{RL}}^2$, which measures the SFT loss increase per RL step. Then the total composite loss $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{SFT}}(\theta_{\text{RL}}^{(2)}) + \lambda \mathcal{L}_{\text{RL}}(\theta_{\text{RL}}^{(2)})$, with trade-off weight $\lambda > 0$, satisfies:*

$$\mathcal{L}_{\text{total}} \geq \mathcal{L}_{\text{SFT}}^* + \frac{\kappa}{2} T_{\text{RL}}^2 - \lambda \eta \bar{g}_{\text{RL}} T_{\text{RL}}, \quad (16)$$

where the first term is the SFT baseline, the second is the accumulated coupling damage (quadratic in T_{RL}), and the third is the RL improvement (linear in T_{RL}). The RL duration that minimizes this lower bound is:

$$T_{\text{RL}}^* = \frac{\lambda \eta \bar{g}_{\text{RL}}}{\kappa} = \frac{\lambda}{\mu_{\text{SFT}} \eta \bar{g}_{\text{RL}}}. \quad (17)$$

Proof After T_{RL} steps of RL with learning rate η , the parameter displacement from $\theta_{\text{SFT}}^{(1)}$ is approximately

$$\|\delta\| = \|\theta_{\text{RL}}^{(2)} - \theta_{\text{SFT}}^{(1)}\| \approx \eta \sum_{t=1}^{T_{\text{RL}}} \|g_{\text{RL}}^{(t)}\| \approx \eta \bar{g}_{\text{RL}} T_{\text{RL}}.$$

Under the PL condition, the quadratic growth property gives the SFT loss increase:

$$\Delta \mathcal{L}_{\text{SFT}} = \mathcal{L}_{\text{SFT}}(\theta_{\text{RL}}^{(2)}) - \mathcal{L}_{\text{SFT}}^* \geq \frac{\mu_{\text{SFT}}}{2} \|\delta\|^2 \geq \frac{\mu_{\text{SFT}}}{2} (\eta \bar{g}_{\text{RL}} T_{\text{RL}})^2 = \frac{\kappa}{2} T_{\text{RL}}^2,$$

where $\kappa = \mu_{\text{SFT}} \eta^2 \bar{g}_{\text{RL}}^2$.

For the RL loss, assuming linear improvement in reward (first-order approximation),

$$\mathcal{L}_{\text{RL}}(\theta_{\text{RL}}^{(2)}) \approx \mathcal{L}_{\text{RL}}(\theta_{\text{SFT}}^{(1)}) - \eta \bar{g}_{\text{RL}} T_{\text{RL}}.$$

The total composite loss lower bound is therefore

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{SFT}}(\theta_{\text{RL}}^{(2)}) + \lambda \mathcal{L}_{\text{RL}}(\theta_{\text{RL}}^{(2)}) \geq \mathcal{L}_{\text{SFT}}^* + \frac{\kappa}{2} T_{\text{RL}}^2 + \lambda (\mathcal{L}_{\text{RL}}(\theta_{\text{SFT}}^{(1)}) - \eta \bar{g}_{\text{RL}} T_{\text{RL}}).$$

Ignoring the constant $\mathcal{L}_{\text{RL}}(\theta_{\text{SFT}}^{(1)})$ term (which does not affect the optimal T_{RL}), we minimize the T_{RL} -dependent part:

$$f(T_{\text{RL}}) = \frac{\kappa}{2} T_{\text{RL}}^2 - \lambda \eta \bar{g}_{\text{RL}} T_{\text{RL}}.$$

Taking the derivative and setting it to zero:

$$\frac{df}{dT_{\text{RL}}} = \kappa T_{\text{RL}} - \lambda \eta \bar{g}_{\text{RL}} = 0 \implies T_{\text{RL}}^* = \frac{\lambda \eta \bar{g}_{\text{RL}}}{\kappa} = \frac{\lambda}{\mu_{\text{SFT}} \eta \bar{g}_{\text{RL}}}.$$

This completes the proof. ■

The inverse dependence on $\eta \bar{g}_{\text{RL}}$ means that larger learning rates or larger RL gradients both prescribe shorter RL training—each step causes more SFT damage via coupling. Moreover, T_{RL}^* is not a fixed hyperparameter: it depends on the landscape geometry (μ_{SFT}) that varies across models, tasks, and training progress.

3.2 Gradient Alignment Bound via Spectral Concentration

The coupling in Theorem 3.1 is driven by gradient misalignment between SFT and RL objectives. We now provide a theoretical bound on this misalignment under a low-rank spectral concentration assumption.

Proposition 2 (Gradient Alignment Bound) *Suppose both gradients have low effective rank when expanded in a common orthonormal basis $\{e_i\}_{i=1}^d$ of $\mathbb{R}^d$: the SFT gradient $g_{\text{SFT}} = \nabla \mathcal{L}_{\text{SFT}}$ concentrates on directions $S \subset \{1, \dots, d\}$ and the RL gradient $g_{\text{RL}} = \nabla \mathcal{J}_{\text{RL}}$ concentrates on directions $R \subset \{1, \dots, d\}$, with $|S|, |R| \ll d$ and $|S \cap R| \ll |S \cup R|$. Then:*

$$|\cos(g_{\text{SFT}}, g_{\text{RL}})| \leq \frac{|S \cap R|}{\sqrt{|S| \cdot |R|}} + O(1/\sqrt{d}), \quad (18)$$

where d is the parameter dimension.

Proof Express the gradients in the chosen orthonormal basis:

$$g_{\text{SFT}} = \sum_{i \in S} \alpha_i e_i + \sum_{i \notin S} \epsilon_i e_i, \quad g_{\text{RL}} = \sum_{j \in R} \beta_j e_j + \sum_{j \notin R} \zeta_j e_j,$$

where S and R are the index sets of dominant directions for each gradient, and the residual coefficients satisfy $\sum_{i \notin S} \epsilon_i^2 \leq \delta_S \|g_{\text{SFT}}\|^2$ and $\sum_{j \notin R} \zeta_j^2 \leq \delta_R \|g_{\text{RL}}\|^2$ for small concentration parameters δ_S, δ_R .

The cosine similarity is:

$$\cos(g_{\text{SFT}}, g_{\text{RL}}) = \frac{\langle g_{\text{SFT}}, g_{\text{RL}} \rangle}{\|g_{\text{SFT}}\| \cdot \|g_{\text{RL}}\|}.$$

The numerator decomposes as:

$$\langle g_{\text{SFT}}, g_{\text{RL}} \rangle = \underbrace{\sum_{i \in S \cap R} \alpha_i \beta_i}_{\text{overlap term}} + \underbrace{\sum_{i \in S \setminus R} \alpha_i \zeta_i + \sum_{j \in R \setminus S} \epsilon_j \beta_j + \sum_{i \notin S, j \notin R} \epsilon_i \zeta_i}_{\text{residual terms}}.$$

By the Cauchy–Schwarz inequality, the overlap term is bounded by:

$$\left| \sum_{i \in S \cap R} \alpha_i \beta_i \right| \leq \sqrt{\sum_{i \in S \cap R} \alpha_i^2} \cdot \sqrt{\sum_{i \in S \cap R} \beta_i^2} \leq \sqrt{|S \cap R|} \cdot \max_{i \in S \cap R} |\alpha_i| \cdot \sqrt{|S \cap R|} \cdot \max_{j \in S \cap R} |\beta_j|.$$

Under the concentration assumption, $\|g_{\text{SFT}}\|^2 \approx \sum_{i \in S} \alpha_i^2$ and $\|g_{\text{RL}}\|^2 \approx \sum_{j \in R} \beta_j^2$, so:

$$\max_i |\alpha_i| \leq \frac{\|g_{\text{SFT}}\|}{\sqrt{|S|}}, \quad \max_j |\beta_j| \leq \frac{\|g_{\text{RL}}\|}{\sqrt{|R|}}.$$

The overlap term is thus bounded by:

$$\left| \sum_{i \in S \cap R} \alpha_i \beta_i \right| \leq \frac{|S \cap R|}{\sqrt{|S|} \cdot \sqrt{|R|}} \cdot \|g_{\text{SFT}}\| \cdot \|g_{\text{RL}}\|.$$

The residual terms contribute at most $O(\sqrt{\delta_S} + \sqrt{\delta_R}) \cdot \|g_{\text{SFT}}\| \cdot \|g_{\text{RL}}\| = O(1/\sqrt{d})$ under standard concentration (by random rotation arguments in high-dimensional spaces (Cai et al., 2013)).

Combining and dividing by $\|g_{\text{SFT}}\| \cdot \|g_{\text{RL}}\|$ yields:

$$|\cos(g_{\text{SFT}}, g_{\text{RL}})| \leq \frac{|S \cap R|}{\sqrt{|S|} \cdot \sqrt{|R|}} + O(1/\sqrt{d}).$$

This completes the proof. ■

This bound shows that when the SFT and RL gradients occupy largely disjoint low-dimensional subspaces, their alignment is necessarily small. The structural mismatch persists regardless of hyperparameter tuning, providing a mechanistic explanation for the near-orthogonality observed empirically and the coupling in Theorem 3.1.

3.3 Non-decoupling Threshold

Theorem 3.1 establishes non-decoupling at the SFT optimum ($\nabla \mathcal{L}_{\text{SFT}} = 0$), where the bound is purely second-order. In practice, SFT may not fully converge, and a first-order term can partially offset the damage. This raises a quantitative question: under what conditions can RL actually improve the SFT loss?

Proposition 3 (Non-decoupling Threshold) *Let $\theta_{\text{SFT}}^{(1)}$ denote the parameter after SFT (not necessarily at the minimum). Let $\bar{d}_{\text{RL}} = -\frac{1}{T_{\text{RL}}} \sum_{t=1}^{T_{\text{RL}}} g_{\text{RL}}^{(t)}$ denote the mean RL update direction (negative mean RL gradient) and let $\bar{g}_{\text{RL}} = \frac{1}{T_{\text{RL}}} \sum_{t=1}^{T_{\text{RL}}} \|g_{\text{RL}}^{(t)}\|$ denote the mean RL gradient norm as in Proposition 1. We parameterize the RL displacement as $\delta = \eta T_{\text{RL}} \bar{d}_{\text{RL}}$. By the second-order Taylor expansion of $\mathcal{L}_{\text{SFT}}$ around $\theta_{\text{SFT}}^{(1)}$:*

$$\Delta \mathcal{L}_{\text{SFT}} = \underbrace{\nabla \mathcal{L}_{\text{SFT}}(\theta_{\text{SFT}}^{(1)})^\top \delta}_{\text{first-order}} + \underbrace{\frac{1}{2} \delta^\top \nabla^2 \mathcal{L}_{\text{SFT}}(\xi) \delta}_{\text{second-order}} \quad (19)$$

for some ξ on the line segment between $\theta_{\text{SFT}}^{(1)}$ and $\theta_{\text{RL}}^{(2)}$. Define the reward-SFT alignment coefficient:

$$\alpha_{\text{RS}} := -\frac{\nabla \mathcal{L}_{\text{SFT}}(\theta_{\text{SFT}}^{(1)})^\top \bar{d}_{\text{RL}}}{\|\nabla \mathcal{L}_{\text{SFT}}(\theta_{\text{SFT}}^{(1)})\| \cdot \|\bar{d}_{\text{RL}}\|}, \quad (20)$$

where $\alpha_{RS} > 0$ means the RL update direction opposes the SFT gradient (potentially improving SFT). Let $\sigma_{\text{eff}}^2 := \bar{d}_{\text{RL}}^\top \nabla^2 \mathcal{L}_{\text{SFT}}(\xi) \bar{d}_{\text{RL}} / \|\bar{d}_{\text{RL}}\|^2$ denote the effective curvature along the RL direction. Then:

$$\Delta \mathcal{L}_{\text{SFT}} < 0 \quad (\text{RL improves SFT}) \quad \iff \quad \alpha_{RS} > \frac{\eta T_{\text{RL}} \sigma_{\text{eff}}^2}{2 \|\nabla \mathcal{L}_{\text{SFT}}(\theta_{\text{SFT}}^{(1)})\|} \cdot \|\bar{d}_{\text{RL}}\| := \tau_{\text{crit}}. \quad (21)$$

Proof Substituting $\delta = \eta T_{\text{RL}} \bar{d}_{\text{RL}}$ into the Taylor expansion (19):

$$\Delta \mathcal{L}_{\text{SFT}} = \eta T_{\text{RL}} \nabla \mathcal{L}_{\text{SFT}}^\top \bar{d}_{\text{RL}} + \frac{\eta^2 T_{\text{RL}}^2}{2} \bar{d}_{\text{RL}}^\top \nabla^2 \mathcal{L}_{\text{SFT}}(\xi) \bar{d}_{\text{RL}}.$$

Using the definition of α_{RS} in (20): $\nabla \mathcal{L}_{\text{SFT}}^\top \bar{d}_{\text{RL}} = -\alpha_{RS} \|\nabla \mathcal{L}_{\text{SFT}}\| \cdot \|\bar{d}_{\text{RL}}\|$. Thus:

$$\Delta \mathcal{L}_{\text{SFT}} = -\eta T_{\text{RL}} \alpha_{RS} \|\nabla \mathcal{L}_{\text{SFT}}\| \cdot \|\bar{d}_{\text{RL}}\| + \frac{\eta^2 T_{\text{RL}}^2}{2} \sigma_{\text{eff}}^2 \|\bar{d}_{\text{RL}}\|^2.$$

For improvement ($\Delta \mathcal{L}_{\text{SFT}} < 0$), the negative first-order term (when $\alpha_{RS} > 0$) must dominate over the positive second-order term:

$$\eta T_{\text{RL}} \alpha_{RS} \|\nabla \mathcal{L}_{\text{SFT}}\| \cdot \|\bar{d}_{\text{RL}}\| > \frac{\sigma_{\text{eff}}^2}{2} \eta^2 T_{\text{RL}}^2 \|\bar{d}_{\text{RL}}\|^2.$$

Dividing both sides by $\eta T_{\text{RL}} \|\nabla \mathcal{L}_{\text{SFT}}\| \cdot \|\bar{d}_{\text{RL}}\|$:

$$\alpha_{RS} > \frac{\sigma_{\text{eff}}^2 \eta T_{\text{RL}} \|\bar{d}_{\text{RL}}\|}{2 \|\nabla \mathcal{L}_{\text{SFT}}\|} = \tau_{\text{crit}}.$$

This completes the proof. ■

The effective curvature σ_{eff}^2 is the Rayleigh quotient of the Hessian along the RL direction—positive (uphill), zero (flat), or negative (downhill). When $\sigma_{\text{eff}}^2 \leq 0$, any positive α_{RS} causes SFT improvement, explaining why RL can improve SFT on tasks with flat loss landscapes. When $\sigma_{\text{eff}}^2 > 0$ (generic case near an SFT minimum), alignment must exceed τ_{crit} to overcome the curvature penalty.

The coefficient α_{RS} is determined by the relationship between RL and SFT objectives. When RL uses a reward substantially different from SFT loss (e.g., human preferences), α_{RS} can be small or negative. When reward partially aligns with SFT (both favor correct outputs), improvement is possible.

4 RL-then-SFT Coupling

In this section, we study the reward of the RL-then-SFT pipeline illustrated in Figure 2(b), where RL is performed on a pretrained base model p_θ followed by SFT. We denote the first RL checkpoint as $p_{\theta_{\text{RL}}^{(1)}}$ and the checkpoint obtained after the additional SFT process as $p_{\theta_{\text{SFT}}^{(2)}}$. We show the irreversibility of the RL and SFT processes.

We first give a weak version of the result. Suppose that SFT starting from an RL policy is a “small” update (e.g., early stopping, regularization, or limited steps). Under this condition, SFT cannot increase the RL reward by more than a constant controlled by the distribution shift budget $B > 0$.

Proposition 4 *Assume that the SFT update does not move too far from the RL policy in average conditional KL:*

$$\mathbb{E}_{\mathbf{x} \sim q} \left[D_{KL} \left(p_{\theta_{SFT}^{(2)}}(\cdot | \mathbf{x}) \| p_{\theta_{RL}^{(1)}}(\cdot | \mathbf{x}) \right) \right] \leq B. \quad (22)$$

Then,

$$\mathbb{E}_{\mathbf{x} \sim q, \mathbf{y} \sim p_{\theta_{SFT}^{(2)}}(\cdot | \mathbf{x})} [r(\mathbf{x}, \mathbf{y})] \leq \mathbb{E}_{\mathbf{x} \sim q, \mathbf{y} \sim p_{\theta_{RL}^{(1)}}(\cdot | \mathbf{x})} [r(\mathbf{x}, \mathbf{y})] + R_{\max} \sqrt{2B}. \quad (23)$$

Proof Fix any $\mathbf{x}$. Let $p_2(\cdot) = p_{\theta_{SFT}^{(2)}}(\cdot | \mathbf{x})$ and $p_1(\cdot) = p_{\theta_{RL}^{(1)}}(\cdot | \mathbf{x})$ be two distributions over $\mathbf{y}$. Since $|r(\mathbf{x}, \mathbf{y})| \leq R_{\max}$, applying Equation (5) with $f(\mathbf{y}) = r(\mathbf{x}, \mathbf{y})/R_{\max}$, the difference in conditional expected reward is controlled by total variation:

$$\left| \mathbb{E}_{\mathbf{y} \sim p_2} [r(\mathbf{x}, \mathbf{y})] - \mathbb{E}_{\mathbf{y} \sim p_1} [r(\mathbf{x}, \mathbf{y})] \right| \leq 2R_{\max} d_{TV}(p_2, p_1), \quad (24)$$

Next, applying Equation (6) yields, for each $\mathbf{x}$,

$$\mathbb{E}_{\mathbf{y} \sim p_2} [r(\mathbf{x}, \mathbf{y})] \leq \mathbb{E}_{\mathbf{y} \sim p_1} [r(\mathbf{x}, \mathbf{y})] + R_{\max} \sqrt{2D_{KL}(p_2 \| p_1)}. \quad (25)$$

Now, taking expectation over $\mathbf{x}$ and applying Jensen’s inequality yields

$$\mathbb{E}_{\mathbf{x} \sim q} [\sqrt{D_{KL}(p_2 \| p_1)}] \leq \sqrt{\mathbb{E}_{\mathbf{x} \sim q} [D_{KL}(p_2 \| p_1)]} \leq \sqrt{B}.$$

Substituting into (25) and averaging over x gives Equation (23). ■

Now, under additional assumptions on the optimality of the first RL checkpoint and bounded shift, we can give a stronger version. The next theorem establishes a non-decoupling property: if the model has already converged under the first RL, so that the RL objective exhibits negligible further reward gain, then the subsequent SFT phase suffers a measurable drop in reward relative to RL-from-scratch. Consequently, the SFT step cannot be decoupled from the earlier RL optimization. In particular, define, for any conditional policy $\pi(\cdot | \mathbf{x})$, the expected reward functional

$$J(\pi) \triangleq \mathbb{E}_{\mathbf{x} \sim q, \mathbf{y} \sim \pi(\cdot | \mathbf{x})} [r(\mathbf{x}, \mathbf{y})]. \quad (26)$$

Then, after the subsequent SFT stage, we would like to show $J(p_{\theta_{SFT}^{(2)}}) < J(p_{\theta_{RL}^{(1)}})$. Hence, RL followed by SFT hurts RL performance, and the two stages cannot be decoupled without incurring a reward deficit.

For this, we assume that the SFT change is bounded and that RL results in a maximizer with quantitative curvature, so any deviation decreases the reward by at least some amount.

Assumption 4 *There exist $0 < a \leq A < \infty$ such that*

$$a \leq \mathbb{E}_{\mathbf{x} \sim q} \left[D_{KL} \left(p_{\theta_{SFT}^{(2)}}(\cdot | \mathbf{x}) \| p_{\theta_{RL}^{(1)}}(\cdot | \mathbf{x}) \right) \right] \leq A. \quad (27)$$

Theorem 4.1 *Under Assumptions 4, the second SFT phase degrades the RL reward obtained in the first RL phase:*

$$J(p_{\theta_{SFT}^{(2)}}) \leq J(p_{\theta_{RL}^{(1)}}) - C_2 \quad (28)$$

for some constant $C_2 > 0$.

Proof By Assumption 4, $p_{\theta_{SFT}^{(2)}}$ is within a region near the local maximizer $p_{\theta_{RL}^{(1)}}$ of $J(\pi)$ bounded by A . Therefore, $\exists \lambda(B) > 0$ such that for every policy π satisfying

$$\mathbb{E}_{\mathbf{x} \sim q} \left[D_{\text{KL}}(\pi(\cdot | \mathbf{x}) \| p_{\theta_{RL}^{(1)}}(\cdot | \mathbf{x})) \right] \leq B \leq A,$$

the following KL-growth condition around $p_{\theta_{RL}^{(1)}}(\cdot | \mathbf{x})$ holds:

$$J(\pi) \leq J(p_{\theta_{RL}^{(1)}}) - \lambda(B) \mathbb{E}_{\mathbf{x} \sim q} \left[D_{\text{KL}}(\pi(\cdot | \mathbf{x}) \| p_{\theta_{RL}^{(1)}}(\cdot | \mathbf{x})) \right].$$

Letting $\pi = p_{\theta_{SFT}^{(2)}}$ yields

$$\begin{aligned} J(p_{\theta_{SFT}^{(2)}}) &\leq J(p_{\theta_{RL}^{(1)}}) - \lambda(B) \mathbb{E}_{\mathbf{x} \sim q} \left[D_{\text{KL}}(p_{\theta_{SFT}^{(2)}}(\cdot | \mathbf{x}) \| p_{\theta_{RL}^{(1)}}(\cdot | \mathbf{x})) \right] \\ &\stackrel{(27)}{\leq} J(p_{\theta_{RL}^{(1)}}) - a\lambda(B), \end{aligned}$$

with $C_2 = a\lambda(B) > 0$. ■

Analogously, under the PL condition on $\mathcal{J}_{RL}$, we obtain a quadratic lower bound on reward degradation.

Theorem 4.2 (PL-based RL Degradation) *Let $\theta_{RL}^{(1)} = \arg \max_{\theta} \mathcal{J}_{RL}(\theta)$ be the RL-optimal parameter. Suppose $\mathcal{J}_{RL}$ satisfies the PL condition with constant $\mu_{RL} > 0$. Then after SFT optimization from this point, the resulting parameter $\theta_{SFT}^{(2)}$ satisfies:*

$$\mathcal{J}_{RL}(\theta_{RL}^{(1)}) - \mathcal{J}_{RL}(\theta_{SFT}^{(2)}) \geq \frac{\mu_{RL}}{2} \|\theta_{SFT}^{(2)} - \theta_{RL}^{(1)}\|^2. \quad (29)$$

Proof The proof is identical to Theorem 3.2 with the roles of SFT and RL reversed. By the PL condition on $\mathcal{J}_{RL}$:

$$\mathcal{J}_{RL}(\theta_{RL}^{(1)}) - \mathcal{J}_{RL}(\theta) \geq \frac{\mu_{RL}}{2} \|\theta - \theta_{RL}^{(1)}\|^2$$

for all θ . Substituting $\theta = \theta_{SFT}^{(2)}$ yields the result. ■

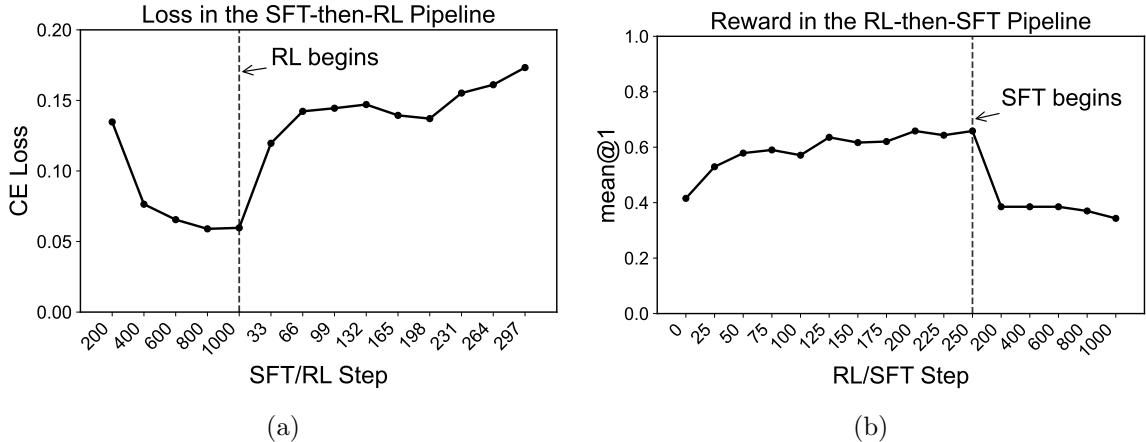


Figure 3: Experimental evidence of coupling. **(a)** SFT-then-RL: SFT loss climbs immediately once GRPO starts and eventually exceeds the base-model baseline. **(b)** RL-then-SFT: reward collapses as soon as SFT begins and falls below the base-model level eventually.

5 Empirical Results

We conduct experiments to verify the non-decoupling properties using the Qwen3-0.6B (Team, 2025) model. We use the Corpus of Linguistic Acceptability (CoLA) dataset (Warstadt et al., 2019), which consists of 10,657 English sentences labeled grammatical or ungrammatical from published linguistics literature. We also prepare a CoLA-style SFT dataset with instruction as the prompt key and output as the response key following the original train/test split. The task we consider is a sentence acceptability classification task in which the model is prompted to judge the grammatical acceptability of a sentence.

We implement both SFT and RL with the VeRL framework (Sheng et al., 2025), running GRPO for RL with a simple reward function that returns $+1$ when the decoded answer matches the ground-truth label and -1 otherwise.

5.1 SFT-then-RL

In this experiment, we first perform SFT on the Qwen3-0.6B base model for 2 complete epochs with the CoLA-style SFT dataset. The resulting checkpoint then serves as the initialization for a reinforcement-learning phase implemented with Group Relative Policy Optimization (GRPO) (Shao et al., 2024).

Figure 3a records the cross-entropy loss on the SFT test set throughout the combined pipeline. When RL begins, the loss increases abruptly and eventually exceeds the value observed for the original base model, a behavior consistent with the bound established in Theorem 3.1.

5.2 RL-then-SFT

In this experiment, we first perform RL with GRPO on the Qwen3-0.6B base model. Once the RL phase converges, we use that model as the initialization for an SFT phase for 2 complete epochs on the same CoLA-style SFT data.

We evaluate the rewards using the same settings as the validation rollout from the RL stage (`temperature` = 0.6, `topp` = 0.95). The reward function is strict on the format in RL training. However, the SFT model can output labels in slightly different formats, therefore assigning -1 even if the answer is semantically correct. To reduce the format sensitivity only in evaluation, we used a robust evaluation which keeps the same scoring rule ($+1$ for correct label, -1 otherwise) while scanning the whole output for “acceptable” / “unacceptable” and use the last occurrence.

We report the `mean@1` reward (one output per prompt) for the entire RL-then-SFT pipeline on the RL test set in Figure 3b. The SFT steps overwrite the RL-tuned behavior and hurt label accuracy, consistent with Theorem 4.1, and a sharp drop in reward can be observed. In particular, the base model (step 0), under robust evaluation, has `mean@1` ≈ 0.385 , i.e. approximately 69.5% accuracy. The final SFT checkpoint has `mean@1` ≈ 0.343 , i.e., approximately 67.2% accuracy, even lower than the base model.

6 Conclusion

Our analysis of the two canonical post-training pipelines shows that supervised fine-tuning and reinforcement learning are inherently coupled: whichever order is adopted, the second stage degrades the performance achieved by the first. We establish this through complementary theoretical lenses: distributional analysis via KL divergence gives nonnegativity of the degradation, while the PL condition yields explicit quadratic bounds that scale with parameter displacement. Our results further reveal that: (i) the coupling damage grows quadratically with RL duration whereas reward improvement grows at most linearly, implying the existence of an optimal stopping point; (ii) gradient near-orthogonality between SFT and RL objectives is a consequence of spectral concentration in different parameter subspaces; and (iii) the non-decoupling threshold identifies precisely when RL can (or cannot) improve SFT performance. Empirical results on the Qwen3-0.6B model corroborate these guarantees, revealing abrupt degradation in either cross-entropy or reward when the transition occurs. We hope that these theoretical insights will inform the development of new training strategies, such as gradient surgery, landscape-aware scheduling, or joint optimization, that better balance memorization and generalization to build more capable models. Recognizing non-decoupling could reduce post-training compute by eliminating destructive RL steps, lowering both carbon emissions and development barriers.

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